



OPEN
Compute Project



OPEN DATA CENTER FOR AI

A Strategic Initiative of the Open Compute Project

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1. Executive Summary

The unprecedented infrastructure build out for Artificial Intelligence and the rise of GW data centers are radically changing how data centers are designed, operated and constructed as dense AI supercomputers blur the lines between IT and facility. To address myriad challenges with delivering this new generation of GW AI data centers at scale, OCP has formed an Open Data Center for AI Initiative leadership team to lead the OCP Community of members and OCP Project teams to address the most pressing challenges specifically:

- Open Data Center Reference Designs
- Energy and Grid Solutions
- Open and Standard Service Level Agreements with Colocation, Neocloud and other 3rd party facility operators
- Telemetry solutions for harmonized IT and Facility operations
- Power estimation methodologies to minimize stranded power

This whitepaper introduces the strategic initiative, its intent and plans in these key areas.

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2. Compliance with Open Compute Project Tenets

2.1 Openness

The OCP Open Data Center for AI initiative is fundamentally about open collaboration in new domains to the community. Bringing OCP's DNA from the Rack and IT infrastructure to facility and energy challenges. Open collaboration on data center reference designs, telemetry and even operation SLAs is already a key part of the initiative.

2.2 Efficiency

Today dense AI data centers are bespoke and deeply tied to individual IT platforms. We seek to foster collaboration to find standards that can simplify and scale AI deployments

2.3 Impact

The initiative is tackling some of the most urgent challenges of the datacenter industry including faster time to power connections and more standardized deployments

2.4 Scale

The leadership team consists of some of the largest AI silicon and hyperscale companies. We anticipate broad adoption of all elements of the initiative.

2.5 Sustainability

Power efficiency is fundamental to all of the priorities outlined in the whitepaper which have a direct impact on sustainability. The fungible data center vision will also result in a longer lifetime for AI facilities and more infrastructure reuse.

3. A new era in data center infrastructure

The rapid evolution of data center IT Infrastructure in the era of artificial intelligence is disrupting the compute landscape from the chip to the grid. Rack power densities have increased by more than an order of magnitude with roadmaps in place today to increase an additional order of magnitude to 1MW in the next few years. GW data center facilities and multi-GW campuses are increasingly common. This new class of GW-scale computing requires a host of new technologies and approaches that are driving the OCP Community to collaborate as never before and pull in new member companies building these solutions at scale.

To address myriad challenges with delivering GW AI data centers at scale, OCP has formed an Open Data Center for AI Initiative leadership team to lead the OCP Community of members and OCP Project teams to address the most pressing challenges. The leadership team has identified 5 priorities for community action:

- Open Data Center Reference Designs
- Energy and Grid Solutions
- Open and Standard Service Level Agreements with Colocation, Neocloud and other 3rd party facility operators
- Telemetry solutions for harmonized IT and Facility operations
- Power estimation methodologies to minimize stranded power

Open Data Center Reference Designs.

During the cloud era the IT rack was the center of focus, but with AI we also need common approaches at the facility level. Ingredient technologies like higher voltage LVDC power (Mt

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Diablo) and Liquid Cooling CDUs (Project Deschutes) have been major efforts in the community in the past year but now taking this a step further, common reference designs incorporating advanced power and cooling for the pod and facility design are sorely needed. The reference designs have three primary goals:

- Fungible facility designs that can support a range of architectures with limited retrofit ensuring flexibility and long life. Fungibility provides the confidence necessary to unlock investment in data center builds more broadly than today.
- Time to first token. A more standard approach to facility designs and common building blocks will improve supply chain integrity and minimize unnecessary debug and delays in deployment. A GW- scale data center is essentially a large computing system and consistency of implementation will avoid substantial technical problems.
- Supply chain alignment and innovation. A standard industry roadmap provides clear demand signals to a deep and diversifying supply chain to increase confidence needed to make capital and R&D investments.

Energy and Grid

Securing power is the primary bottleneck driving data center site selection and development. GW scale data centers are so consequential to the grid that they must move beyond acting as a passive load, to perform services as an integrated part of the grid, supporting utilities in delivering a reliable and affordable service for all.

Load Flexibility

One of the most exciting opportunities is to capitalize on the very low utilization of the grid outside of peak demand cycles. The grid is designed to ensure reliability even during extreme

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weather events and peaks. The vast majority of the time the grid is utilized at 50% or less. If peaks comprising just 0.25% of annual hours could be avoided 75GW of capacity would be available in North America alone according to a Duke University study (Noriss et al). If data centers can utilize this latent capacity not only will deployments be accelerated but ratepayers will actually benefit from the data center investment as utilities better monetize existing capacity.

To pursue this opportunity OCP and the Electric Power Research Institute are collaborating on how to implement Flexibility between the grid and the data center. Much work is needed to define parameters, data flows and contract requirements. Not all nodes, regions and utilities needs are the same. Similarly, not all data centers have the same flexible capabilities.

Standard technical specifications

The large power loads of AI Data Centers involve significant fluctuations and can impact grid reliability. A number of key technical issues with grid connection could benefit from more standard approaches, best practices documents, and reference designs. Including Low Voltage Ride Through (where a dip in grid voltage cannot result in a drop in load) or maintaining load continuity through synchronous AI training workloads. In 2026 the regulatory landscape is expected to shift significantly with connection requirements for data centers being defined in many geographies.

On Site Generation and Energy Storage

Adoption of on site generation and energy storage is accelerating, and the industry is experiencing a steep learning curve in adopting these technologies for the AI Data Center. We expect the community to collaborate on data center- specific BESS (Battery Energy Storage Systems), microgrids, and strategies for load matching with on- site turbines.

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Open Service Level Agreements (Open SLA)

Modern data centers often involve multiple parties working together. The data center facility owner may differ from the IT equipment owner which may differ still from the company running the IT workload. While SLAs for 3rd party operators in the cloud era were relatively straightforward with conventional air cooling and power delivery, the AI Data Center involves a tighter interaction between the IT equipment and advanced cooling and power systems of the facility. These Service Level Agreements can have unintended consequences for energy efficiency, cost, water use and capacity. The OCP Community has taken up the issue with contributions from key hyperscale companies to develop an open set of best practices for these agreements with the opportunity to simplify, standardize and continuously improve how companies can work together to achieve better overall outcomes.

Telemetry

“Telemetry, the automated collection and analysis of metrics spanning power, thermal, and mechanical systems, is the key to unlocking the next generation of data center operation.” (Sharp et al) With the use of AI techniques, we can envision a host of powerful applications to sense and manage the GW data center more effectively. Benefits include minimized power and water consumption, predictive failure analysis and improved reliability. With such a large number of companies and devices present in the data center, current standards and common approaches need to be enhanced. Traditionally telemetry and manageability between facilities (“OT”) and IT equipment is limited for security reasons but opportunities may exist to enhance operations. New categories of equipment that sit between traditional IT and OT are now also important to consider such as Cooling Distribution Units and LVDC power distribution. New Facility Telemetry standards have already been developed by the OCP Community which we intend to extend to these other domains.

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Power Estimation Methodologies.

A surprising fact is that the amount of equipped power needed for an AI Data Center supporting a given number of GPUs may vary significantly across the industry. Actual consumed power can be significantly affected by operating points, workload and software, but also cooling and power delivery systems. Even SLA agreement terms can have unintended consequences that increase power consumption to achieve artificial constraints. New architectures and silicon components also may have significant variability until late in the development cycle and well after data center architectural decisions have to be made.

The OCP Community is bringing together experts from a variety of companies to share best practices and drive continuous improvement. These are expected to be especially valuable for 3rd-party data center operators.

Call to Action

The Open Compute Project has formed the Open Data Center for AI leadership team with representatives from key hyperscale and silicon providers to address these challenges. Our approach is to leverage existing community projects where possible and initiate new ones where needed. Progress to date includes:

- Contributions of the initial OCP Open Data Center Reference Design from Google with collaborations now expected from the broader community. We expect future contributions of other similar reference designs and community action to drive consolidation
- OCP and EPRI have signed an MOU initiating collaboration on Flexible Data Centers and Connection requirements

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- Formation of an Energy OCP Project team focuses on data center to grid and energy related collaboration.
- Engagement with broad set of colocation and neocloud data center providers for feedback on initiatives including reference designs and the Open SLA

The AI revolution is an exciting moment reimagining the future of the data center. The Open Compute Project is embracing this opportunity to bring an even larger community together to solve critical challenges and accelerate the industry for the benefit of everyone.

4. References

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5. License

5.1 Creative Commons

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6. About Open Compute Foundation

The Open Compute Project (OCP) brings at-scale innovations and hyperscaler best practices to all, spanning technology domains from the data center to the edge, and the technology stack from silicon, to systems, to site facilities and services. The international OCP Community is made up of organizations and people from hyperscale and tier-2 cloud data center operators, communications providers, colocation providers, diverse enterprises, and technology vendors. The OCP Foundation fosters collaboration within the Community to tackle market challenges in an array of OCP Projects that create open contributions such as specifications, designs, and more. The OCP Solution Provider Program then recognizes the application of those contributions in compliant products, solutions, and data center facilities and services with designations like OCP Inspired, OCP Accepted and OCP Ready. With the tenets of openness, impact, efficiency, scale and sustainability, the OCP engages and educates thousands of engineers every year through many events and webinars. Across many initiatives the OCP Foundation and Community are meeting the market today and shaping the future.

Learn more at: www.opencompute.org.